Operation Management

BITS Pilani
Pilani Campus

Abhishek Maiti

BITS Pilani
Pilani Campus

<MBAZG526/QMZG526, Operation Management>
Lecture No. 7

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

Layout Strategies

- The objective of layout strategy is to develop an effective and efficient layout that will meet the firm's competitive requirements .

Layout Strategies

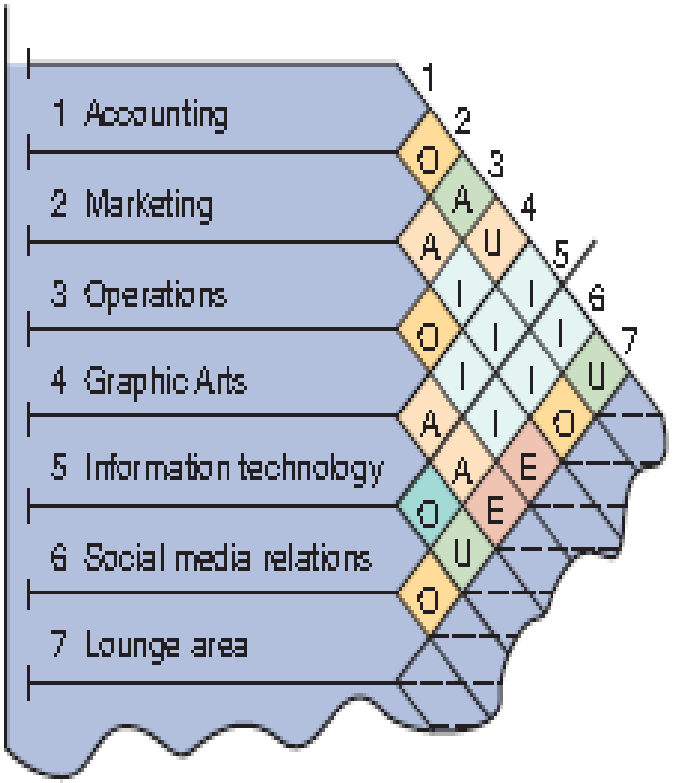
TABLE 9.1 Layout Strategies

	OBJECTIVES	EXAMPLES
Office	Locate workers requiring frequent contact close to one another	Allstate Insurance Microsoft Corp.
Retail	Expose customer to high-margin items	Kroger's Supermarket Walgreens Bloomingdale's
Warehouse (storage)	Balance low-cost storage with low cost material handling	Federal-Mogul's warehouse The Gap's distribution center
Project (fixed position)	Move material to the limited storage areas around the site	Ingall Ship Building Corp. Trump Plaza Pittsburgh Airport
Job shop (process oriented)	Manage varied material flow for each product	Arnold Palmer Hospital Hard Rock Cafe Olive Garden
Work cell (product families)	Identify a product family, build teams, cross-train team members	Hallmark Cards Wheeled Coach Ambulances
Repetitive/continuous (product oriented)	Equalize the task time at each workstation	Sony's TV assembly line Toyota Scion

Source: Heizer, Render, Muckson

Office Layout

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



Code	CLOSENESS
A	Absolutely necessary
E	Especially important
I	Important
O	Ordinary OK
U	Unimportant
X	Not desirable

Source: Heizer, Render, Muckson

Here are five versions of the office layout.

Managers and architects have pondered how to design an office to encourage productivity for more than 100 years. In the early 20th century, large offices resembled factories (see the photo of the Jack Lemmon film *The Apartment*, where clerical workers sat in long rows, often performing repetitive tasks).

Starting in the 1960s, layouts changed to foster teamwork where managers and support staff sat together, and groupings were geared toward specific tasks.

With computers, more individual work was possible and the "Cube Farm" era became ubiquitous through the '80s and '90s. An office full of high-walled cubicles offered both an open environment and personal office space.

By the turn of the century, looking for innovation and creativity to recruit and inspire college grads, technology firms created the "fun" office. Bright, casual, open office spaces, with amenities such as beanbag chairs, foosball tables, and coffee bars became the fad.

The buzzwords today are serendipity and collaboration, as companies design office space to engineer encounters between employees. Steve Jobs designed his Pixar headquarters with the cafeteria and bathrooms in a central atrium away from work areas to encourage intermingling and collaboration. Skype achieves similar goals with open lounges.

Sources: Wall Street Journal (April 28, 2014); *USA Today* (Feb. 28, 2013); and *Harvard Business Review* (Oct. 2014).

Retail Layout

- The main objective of retail layout is to maximize profitability per square foot of floor space.
- An additional, and somewhat controversial, issue in retail layout is called slotting.
- Slotting fees are fees manufacturers pay to get their goods on the shelf in a retail store or supermarket chain.

Maximize Profitability per sft of Floor Space

- Use prominent locations for high-impulse and high-margin items.
- Distribute what are known in the trade as "power items" to both sides of an aisle.
- Use end-aisle locations because they have a very high exposure rate.
- Convey the mission of the store by carefully selecting the position of the lead-off items.

Service Space

To provide a good service layout, a firm considers three elements:

1. Ambient conditions: Fine Dining Restaurant.

2. Spatial layout and functionality: Best Buy, More, Big Bazaar, Spencer.

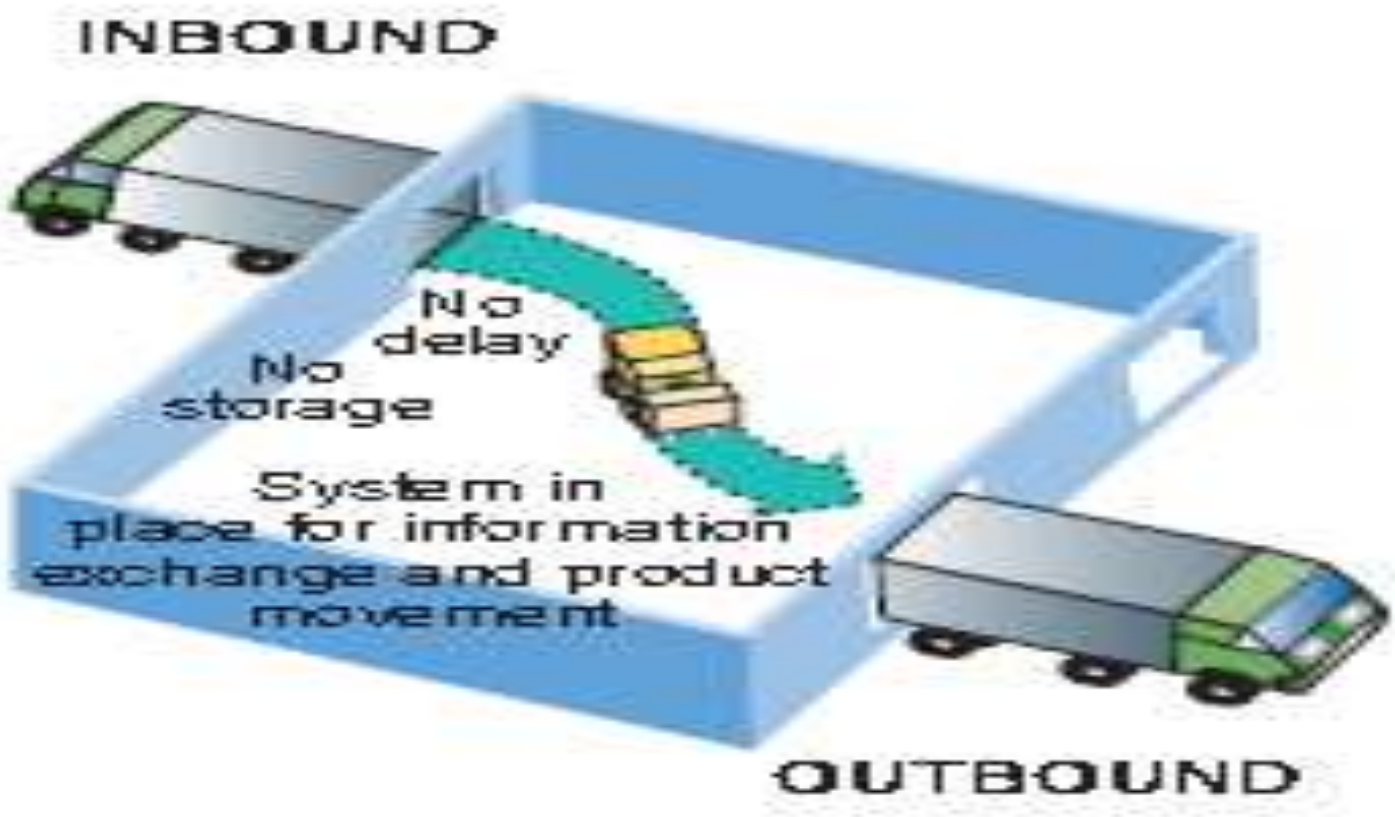
3. Signs, symbols, and artefacts: Hard Rock Café.

Warehouse and Storage Layout

- A design that attempts to minimize total cost by addressing trade-offs between space and material handling.
- In warehouse layout, we want to maximize use of the whole building—from floor to ceiling.
- An important component of warehouse layout is the relationship between the receiving/unloading area and the shipping/loading area.
- Stocking Options : Cross Docking, Random Stocking and Customizing.

Cross Docking

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



Source: Heizer, Render, Muckson

Random Stocking

- Used in warehousing to locate stock wherever there is an open location. Suitable for accessories.
- Computerized random stocking systems (Read Text book).

Customizing

- Using warehousing to add value to a product through component modification, repair, labelling, and packaging.
- Patagonia.

Fixed Position Layout

- A system that addresses the layout requirements of stationary projects.
- In a fixed-position layout , the project remains in one place, and workers and equipment come to that one work area.
- Limited space.
- At different stages of a project, different materials are needed.
- The volume of materials needed is dynamic.

Fixed Position Layout

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

Here are three versions of the fixed-position layout.

Craig Ruttle/AP Images

A house built via traditional fixed-position layout would be constructed onsite, with equipment, materials, and workers brought to the site. Then a "meeting of the trades" would assign space for various time periods. However, the home pictured here can be built at a much lower cost. The house is built in two movable modules in a factory. Scaffolding and hoists make the job easier, quicker, and cheaper, and the indoor work environment aids labor productivity.

Dick Blume/The Image Works

A service example of a fixed-position layout is an operating room; the patient remains stationary on the table, and medical personnel and equipment are brought to the site.

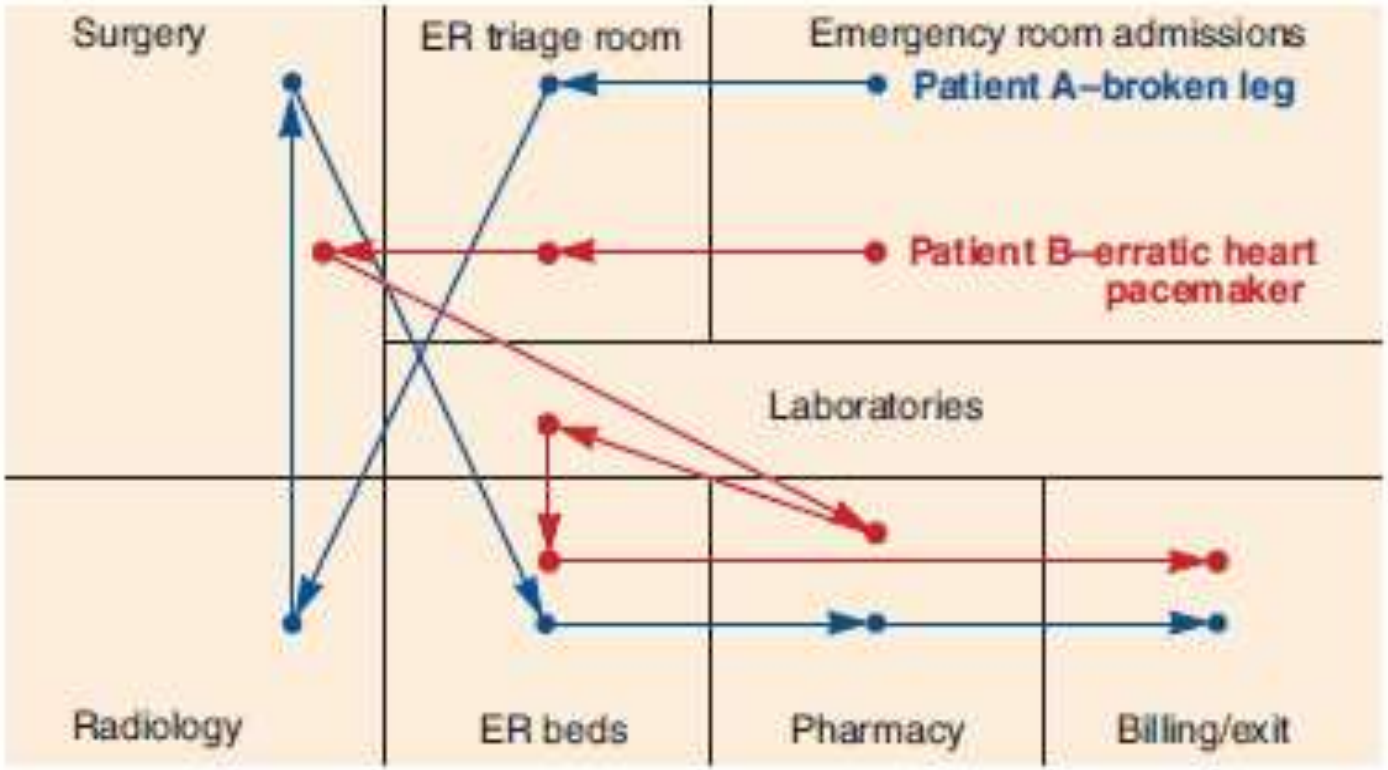
Tim Aubry/Reuters/Corbis

In shipbuilding, there is limited space next to the fixed-position layout. Shipyards call these loading areas platens, and they are assigned for various time periods to each contractor.

Source: Heizer, Render, Muckson

Process Oriented Layout

- Different machines are grouped together.
- A big advantage of process-oriented layout is its flexibility in equipment and labour assignments.
- The disadvantages of process-oriented layout is orders take more time to move through the system because of difficult scheduling, changing setups, and unique material handling.


Surgery
ER triage room
Emergency room admissions
Patient A–broken leg
Patient B–erratic heart pacemaker
Laboratories
Radiology
ER beds
Pharmacy
Billing/exit

Any Question

Class Exercise

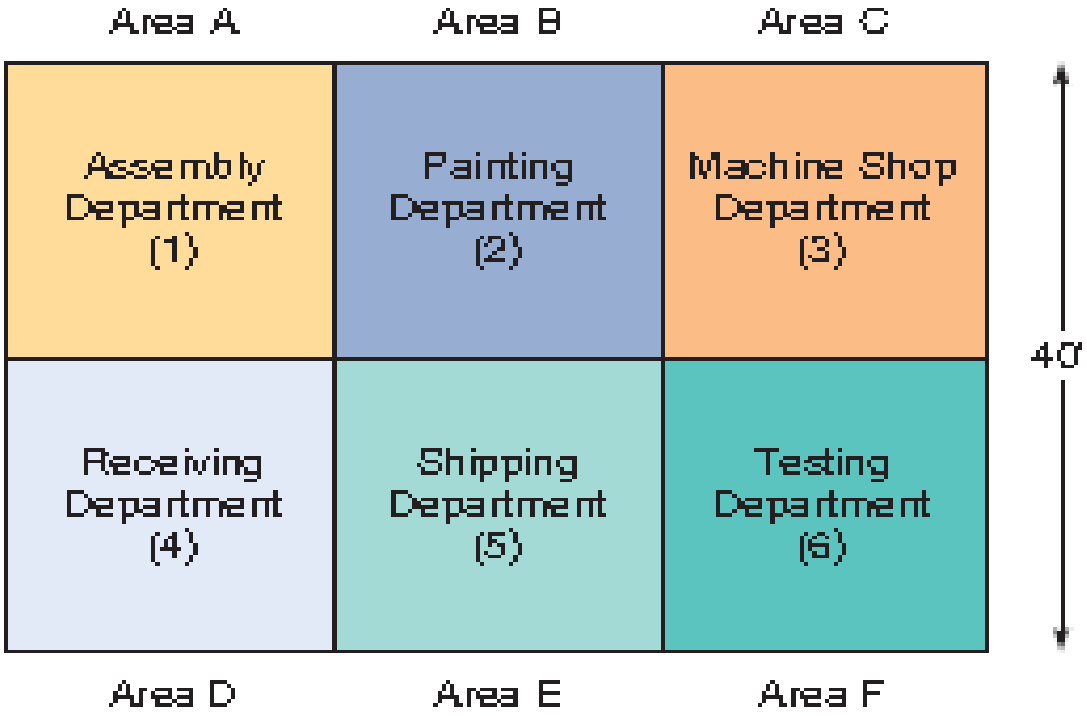
- Walters Company management wants to arrange the six departments of its factory in a way that will minimize interdepartmental material-handling costs. What type of Layout?

Number of loads per week

Department	Assembly (1)	Painting (2)	Machine Shop (3)	Receiving (4)	Shipping (5)	Testing (6)
Assembly (1)		50	100	0	0	20
Painting (2)			30	50	10	0
Machine Shop (3)				20	0	100
Receiving (4)					50	0
Shipping (5)						0
Testing (6)						

Source: Heizer, Render, Muckson

Solution

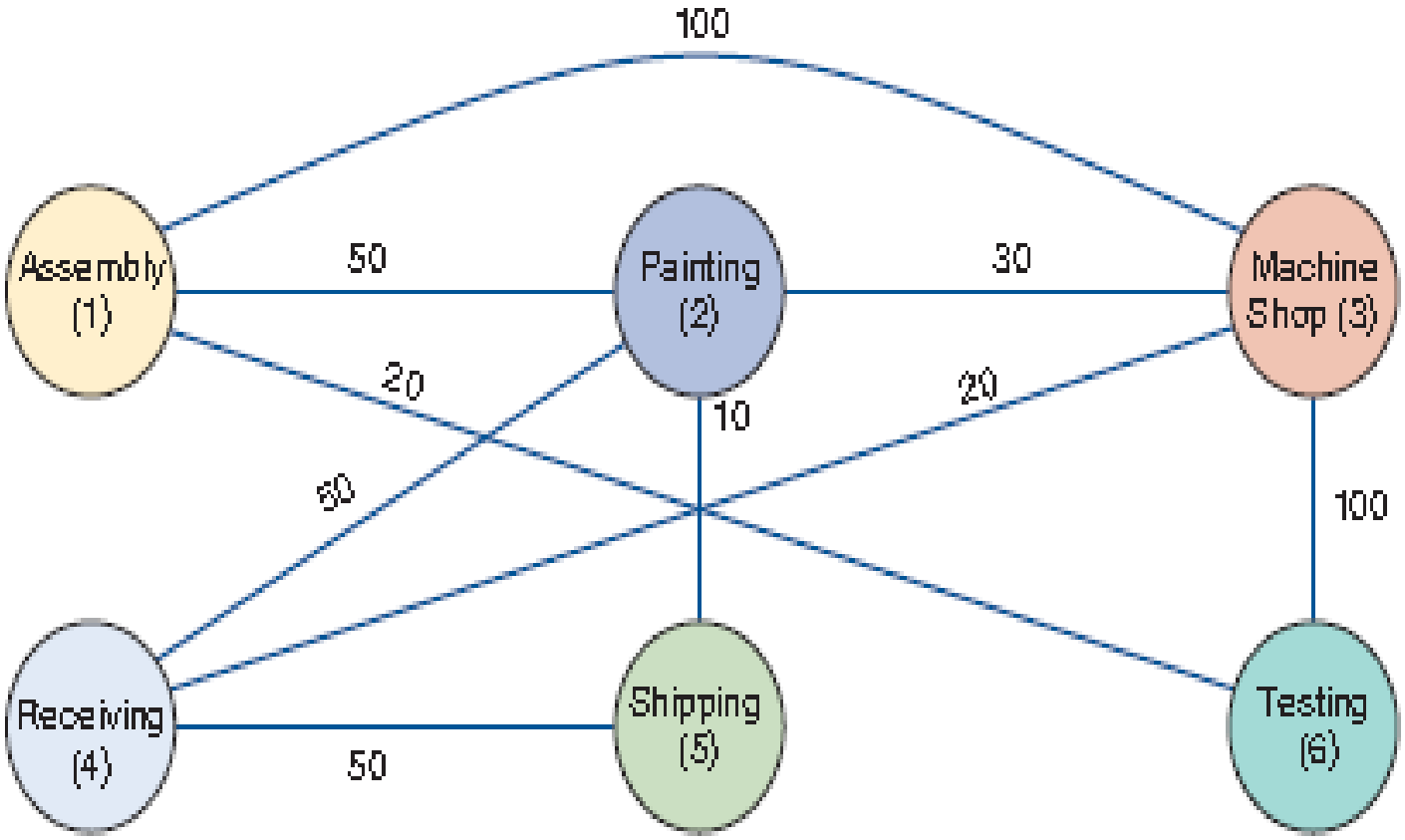


- For this problem, Walters Company assumes that a forklift carries all interdepartmental loads.
- The cost of moving one load between adjacent departments is estimated to be $1. Moving a load between nonadjacent departments costs $2.

Source: Heizer, Render, Muckson

Initial Schematic Diagram

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



Source: Heizer, Render, Muckson

Cost Calculation

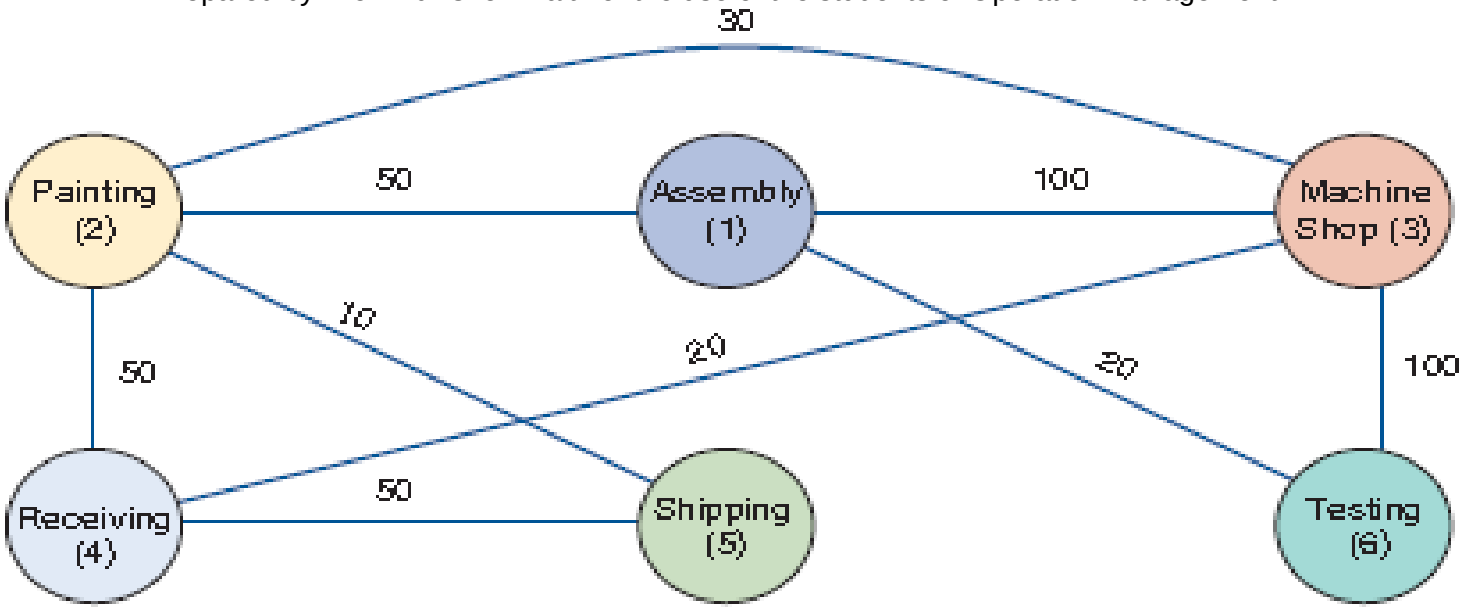
$$\text{Cost} = \sum_{i=1}^{n}\sum_{j=1}^{n} X_{ij} C_{ij}$$

$$\begin{aligned}
\text{Cost} = {} & \underset{(1 \text{ and } 2)}{\$50} + \underset{(1 \text{ and } 3)}{\$200} + \underset{(1 \text{ and } 6)}{\$40} + \underset{(2 \text{ and } 3)}{\$30} + \underset{(2 \text{ and } 4)}{\$50} \\
& + \underset{(2 \text{ and } 5)}{\$10} + \underset{(3 \text{ and } 4)}{\$40} + \underset{(3 \text{ and } 6)}{\$100} + \underset{(4 \text{ and } 5)}{\$50} \\
= {} & \$570
\end{aligned}$$

Source: Heizer, Render, Muckson

Modified Schematic Diagram & Cost Calculation

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



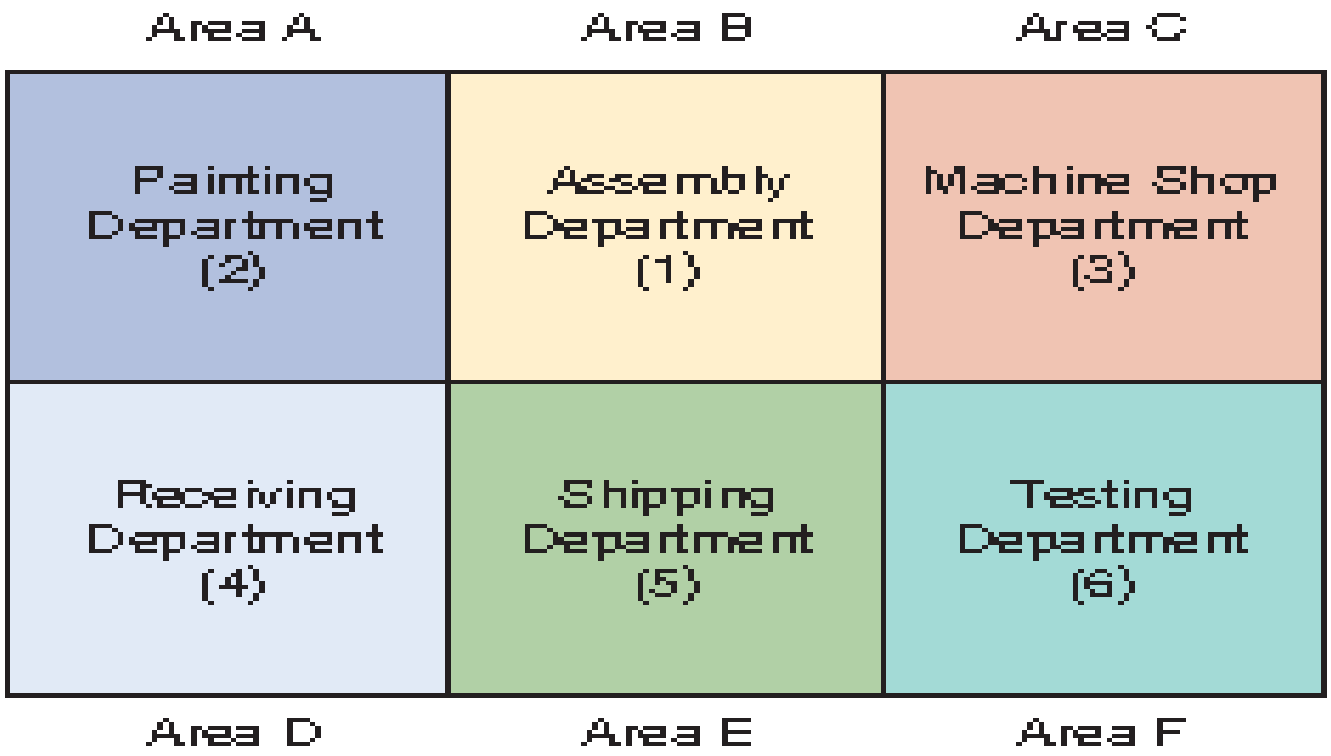
$$\text{Cost} = \underset{(1 \text{ and } 2)}{\$50} + \underset{(1 \text{ and } 3)}{\$100} + \underset{(1 \text{ and } 6)}{\$20} + \underset{(2 \text{ and } 3)}{\$60} + \underset{(2 \text{ and } 4)}{\$50}$$

$$+ \underset{(2 \text{ and } 5)}{\$10} + \underset{(3 \text{ and } 4)}{\$40} + \underset{(3 \text{ and } 6)}{\$100} + \underset{(4 \text{ and } 5)}{\$50}$$

$$= \$480$$

Source: Heizer, Render, Muckson

Final Layout

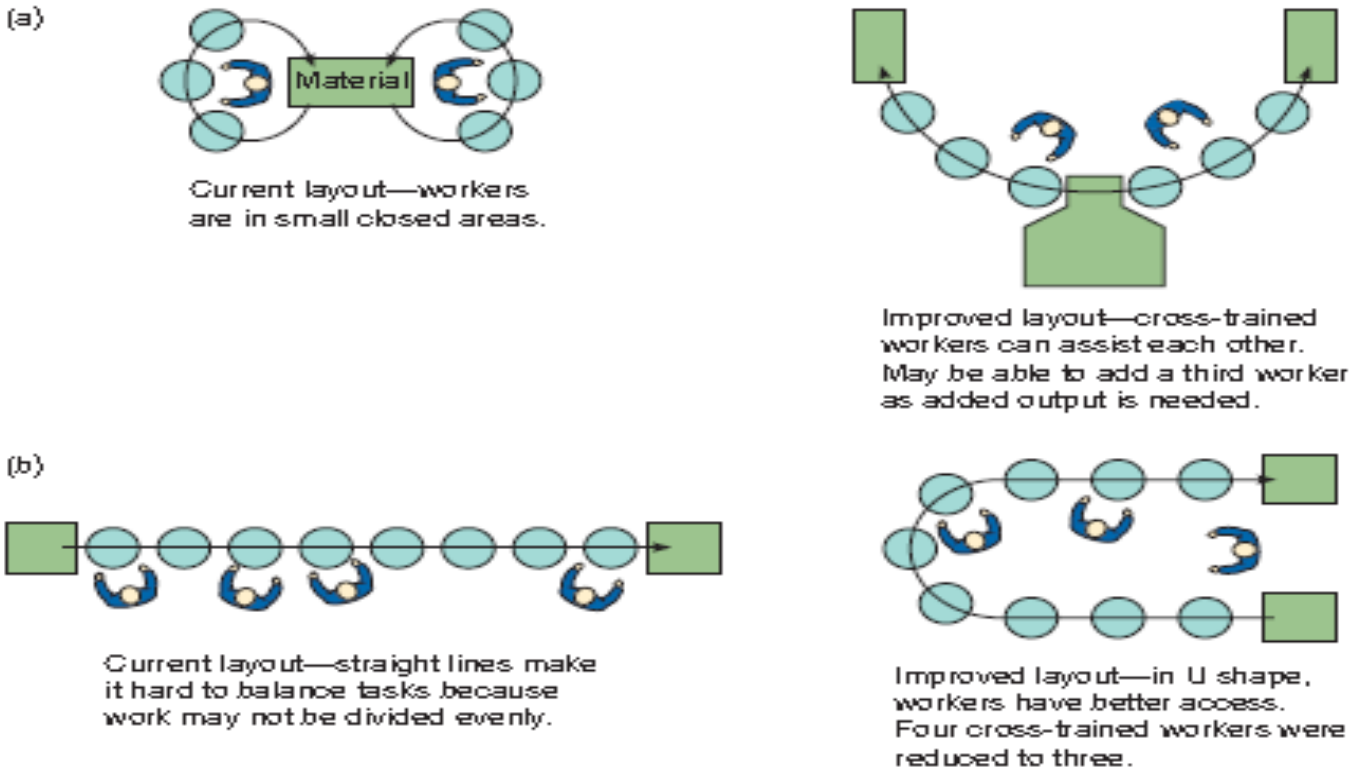


Source: Heizer, Render, Muckson

Any Question

Work Cells

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



Source: Heizer, Render, Muckson

Cellular Production Requirements

- Identification of families of products, often through the use of group technology codes or equivalents.
- A high level of training, flexibility, and empowerment of employees.
- Being self-contained, with its own equipment and resources.
- Testing (poka-yoke) at each station in the cell.

Advantages of Work Cells

- Work cells have at least five advantages over assembly lines and focused process facilities: (1) because tasks are grouped, inspection is often immediate; (2) fewer workers are needed; (3) workers can reach more of the work area; (4) the work area can be more efficiently balanced; and (5) communication is enhanced.

Work Cell Calculation

$$Takt\ Time = \frac{Total\ Work\ time\ available}{Customer\ Demand}$$

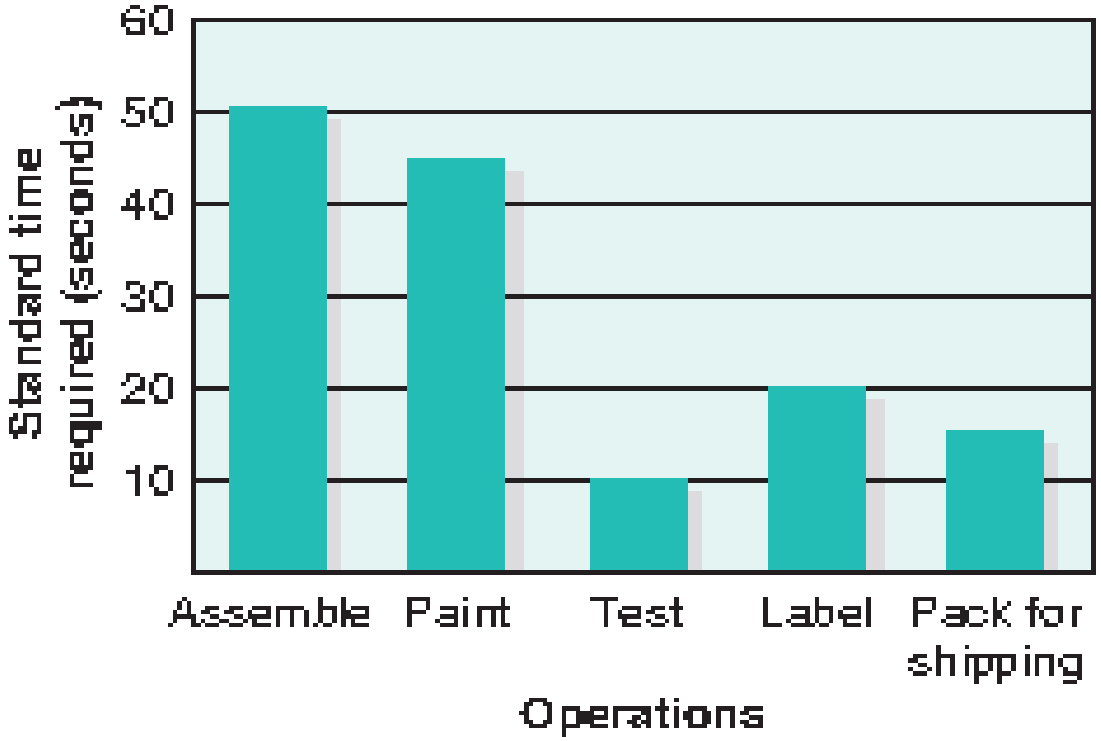
$$Workers\ requird = \frac{Total\ Operation\ time}{Takt\ time}$$

Class Exercise

- Stephen Hall's company in Dayton makes auto mirrors. The major customer is the Honda plant nearby.
- Honda expects 600 mirrors delivered daily, and the work cell producing the mirrors is scheduled for 8 hours.
- Hall wants to determine the takt time and the number of workers required.

Class Exercise

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



How Will You Arrange the Workers?

Source: Heizer, Render, Muckson

Repetitive and Product-Oriented Layout

- Product-oriented layouts are organized around products or families of similar high-volume, low-variety products.
- Two types of a product-oriented layout are fabrication and assembly lines.
- The fabrication line builds components, such as automobile tires or metal parts.
- An assembly line puts the fabricated parts together.

Assembly Line Balancing

- Obtaining output at each workstation on a production line so delay is minimized.
- Management also needs to know the *precedence relationship* among the activities.

Class Exercise

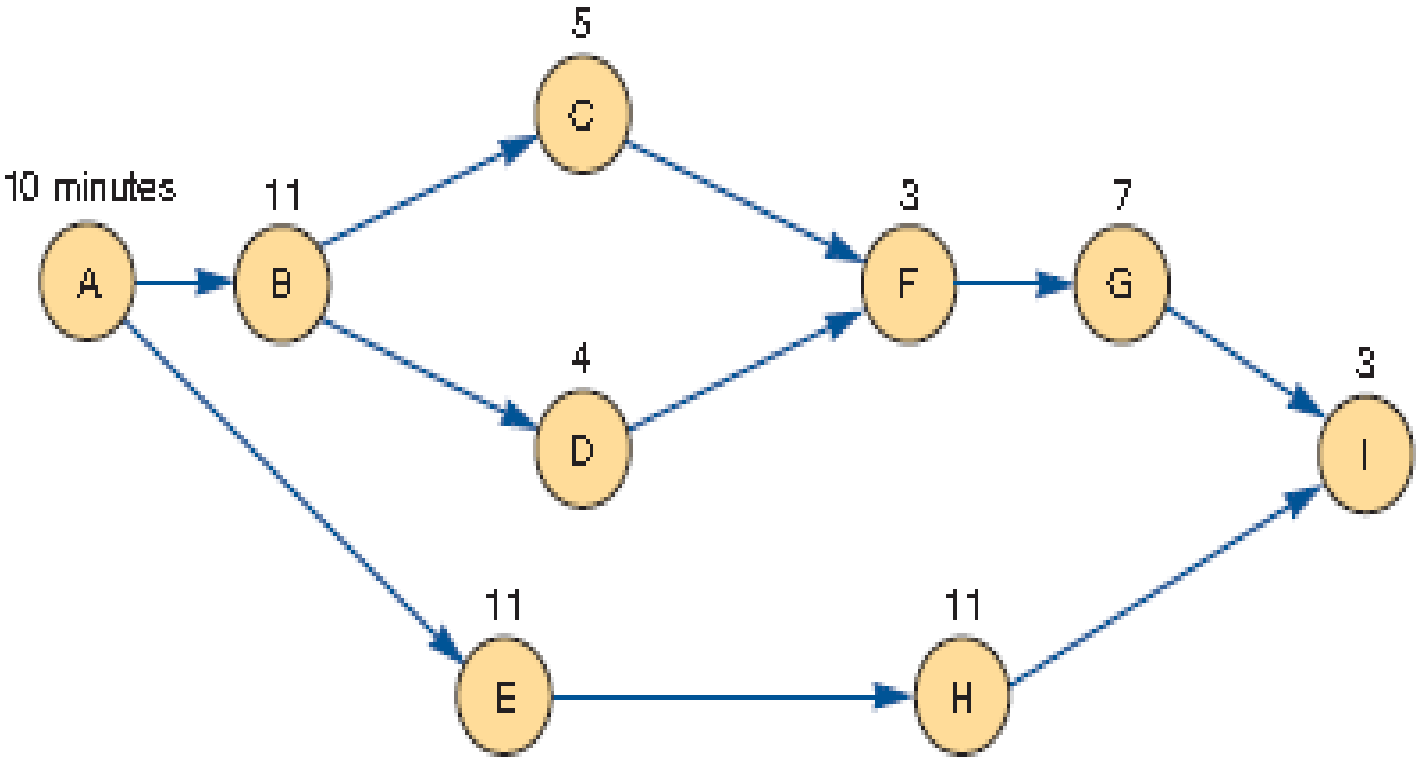
- Boeing wants to develop a precedence diagram for an electrostatic wing component that requires a total assembly time of 65 minutes.

TABLE 9.2 Precedence Data for Wing Component

TASK	ASSEMBLY TIME (MINUTES)	TASK MUST FOLLOW TASK LISTED BELOW	
A	10	—	This means that tasks B and E cannot be done until task A has been completed.
B	11	A	
C	5	B	
D	4	B	
E	11	A	
F	3	C, D	
G	7	F	
H	11	E	
I	3	G, H	
	Total time 65		

Source: Heizer, Render, Muckson

Solution



Source: Heizer, Render, Muckson

Theory

- Once we have constructed a precedence chart, we turn to the job of grouping tasks into job stations so that we can meet the specified production rate.

- $$Cycle\ time = \frac{Production\ time\ avaialble\ per\ day}{Units\ required\ per\ day}$$

- $$Minimum\ no.\ of\ workstation = \frac{\Sigma\ time\ for\ task}{Cycle\ time}$$

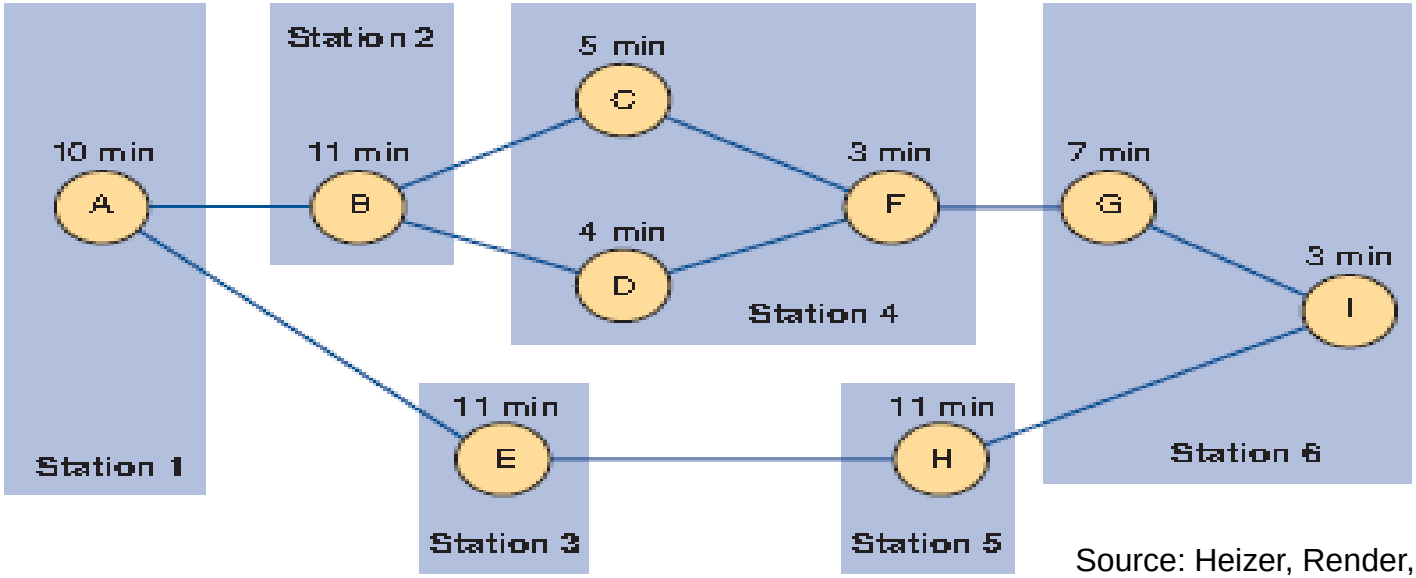
Class Exercise

- On the basis of the precedence diagram and activity times given in previous class exercise , Boeing determines that there are 480 productive minutes of work available per day.
- Furthermore, the production schedule requires that 40 units of the wing component be completed as output from the assembly line each day.
- What is the total Idle Time?

Solution

- $\text{Cycle time} = \dfrac{480 \text{ minutes}}{40 \text{ Units}} = 12 \text{ minutes/ unit}$

- $\text{Minimum no. of workstation} = \dfrac{65}{12} = 5.42 = \sim 6$



Source: Heizer, Render, Muckson

Heuristic Model

TABLE 9.3 Layout Heuristics That May Be Used to Assign Tasks to Workstations in Assembly-Line Balancing

1. *Longest task (operation) time*	From the available tasks, choose the task with the largest (longest) time.
2. *Most following tasks*	From the available tasks, choose the task with the largest number of following tasks.
3. *Ranked positional weight*	From the available tasks, choose the task for which the sum of the times for each following task is longest. (In Example 4 we see that the ranked positional weight of task C = 5(C) + 3(F) + 7(G) + 3(I) = 18, whereas the ranked positional weight of task D = 4(D) + 3(F) + 7(G) + 3(I) = 17; therefore, C would be chosen first, using this heuristic.)
4. *Shortest task (operations) time*	From the available tasks, choose the task with the shortest task time.
5. *Least number of following tasks*	From the available tasks, choose the task with the least number of subsequent tasks.

- Best solution is smallest number of workstations and highest efficiency.
- Remember, however, that although heuristics provide solutions, they do not guarantee an optimal solution

$$\text{Efficiency} = \frac{\sum \text{Task times}}{(\textit{Actual}\ \text{number of workstations}) \times (\text{Largest assigned cycle time})}$$

$$\text{Efficiency} = \frac{65\ \text{minutes}}{(6\ \text{stations}) \times (12\ \text{minutes})}$$

$$= 90.3\ \%$$

Any Question

Further Reading / References

1. **Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 9**

THANK YOU